SBT-DF204 – Computer Forensics Case Study 3

Investigating Memory Evidence: Suspected Illegal File Transfer

Field	Details
Student Name	Adekunle Ogunyemi
Course	SBT-DF204 – Computer Forensics Case Studies
Assessment	Case Study 3 – Individual Memory Forensics Investigation
Case Title	Investigating Suspected Illegal File Transfer From Memory Evidence
Operating System	Kali Linux Forensic Tool Volatility 2.6.1
Evidence File	memdumpWin7.mem

1. Executive Summary

This case study involved the investigation of a Windows 7 memory image to determine whether a confidential document was transferred without authorisation. The purpose of the investigation was to examine the supplied memory dump and determine whether the available evidence supported the suspected unauthorised transfer of a file named secret_file.docx.

The investigation was carried out using Kali Linux and Volatility 2.6.1. The supplied memdumpWin7.mem file was first checked and preserved

before analysis. A working copy was created so that the original evidence would not be used directly during the investigation.

The memory analysis showed that the active user profile was IEUser. The most important finding came from the command history and console output plugins. The memory contained evidence that the user ran the command copy secret_file.docx F: from the Downloads folder, and the console output confirmed that the operation was successful with the message "1 file(s) copied."

Further registry analysis of the MountedDevices key proved that the F: drive was a physical USB Mass Storage Device (USB flash drive). File Explorer history (ShellBags) confirmed that the user had browsed the Downloads folder and the F: drive.

Based on the evidence examined, the memory image strongly connects the IEUser account to the unauthorized copy of secret_file.docx to a USB device. However, the evidence does not prove with absolute certainty that a specific physical person was operating the computer at that exact time, as user accounts can be shared or left logged in.

2. Case Background

Investigators obtained a Windows 7 memory image from a workstation after a confidential document may have been transferred without authorization. The inquiry focuses on whether browser activity, a command prompt, a removable device, and a user profile form a defensible sequence of events.

The main purpose of this investigation was therefore to examine the memory image and determine:

- What operating system profile and image time are supported.
- What process and network evidence relates to the transfer.
- What the command history shows.
- What removable-media evidence is present.

- What File Explorer evidence supports the sequence.
- What conclusion can be defended from the available evidence.

3 Objectives of the Investigation

The objectives of this investigation were:

- To preserve the supplied memory image and verify its integrity.
- To identify the correct Volatility profile.
- To identify relevant processes and network connections.
- To recover the command prompt history and console output.
- To identify the drive letter and device type of the destination drive.
- To examine File Explorer traces that could support the sequence.
- To establish a timeline of the relevant activity in UTC.
- To provide a conclusion based only on the evidence available in the memory image.

Forensic Environment
The investigation was performed using the following environment:

Item	**Details**
Operating System	Kali Linux
Main forensic tool	Volatility 2.6.1
Evidence format	Raw Memory Image (.mem)
Evidence file	memdumpWin7.mem
Working file	working_memdumpWin7.mem
Main plugin examined	cmdscan, consoles, printkey
Main service examined	Windows Command Prompt (cmd.exe)

The analysis was performed on a working copy of the memory image rather than making changes to the original evidence file.

4 Evidence Acquisition and Integrity

4.1 Evidence File

The supplied evidence file was: *memdumpWin7.mem*

The file was stored in the evidence directory:
SBT-DF204-Case_Study_3/evidence/memdumpWin7.mem

The Kali Linux terminal showed the file size as 10,736,762,288 bytes.

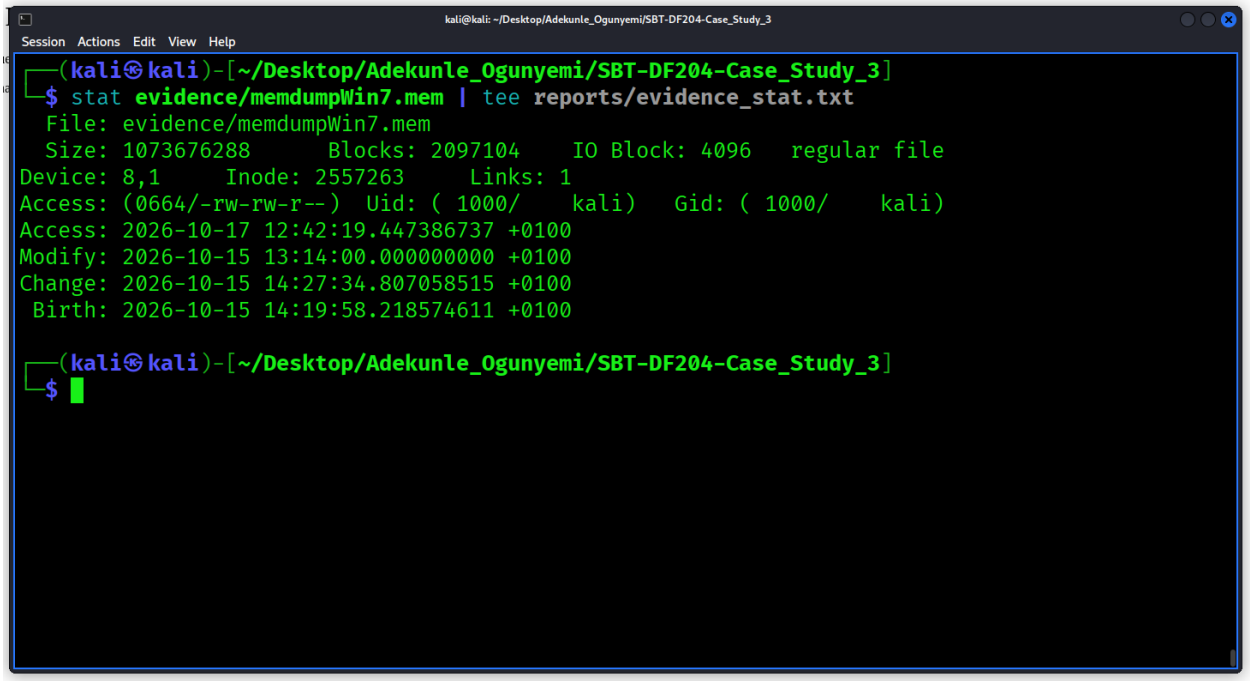


Figure 1: Evidence file memdumpWin7.mem and its recorded file size

The screenshot shows the terminal command used to list the evidence file, including the file name and size.

4.2 SHA-256 Verification

The SHA-256 value of the evidence file was calculated using:
sha256sum evidence/memdumpWin7.mem
sha256sum working/working_memdumpWin7.mem

The calculated hash was:
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e

A working copy was also created and hashed. The SHA-256 value of the original file and the working copy matched.

This shows that the working copy was not changed during the copying process.

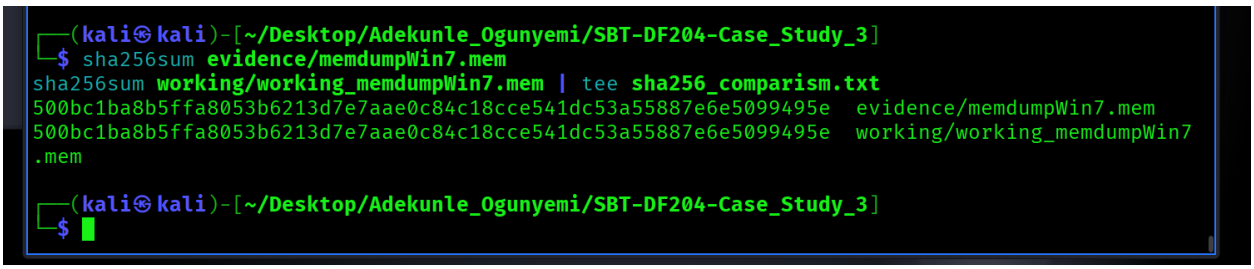


Figure 2: SHA-256 comparison of the original evidence file and the working copy

4.3 Creation of the Working Copy

The working copy was created and verified before analysis. The working copy was then used for all Volatility commands:

python2 volatility/vol.py -f working/working_memdumpWin7.mem [plugin]

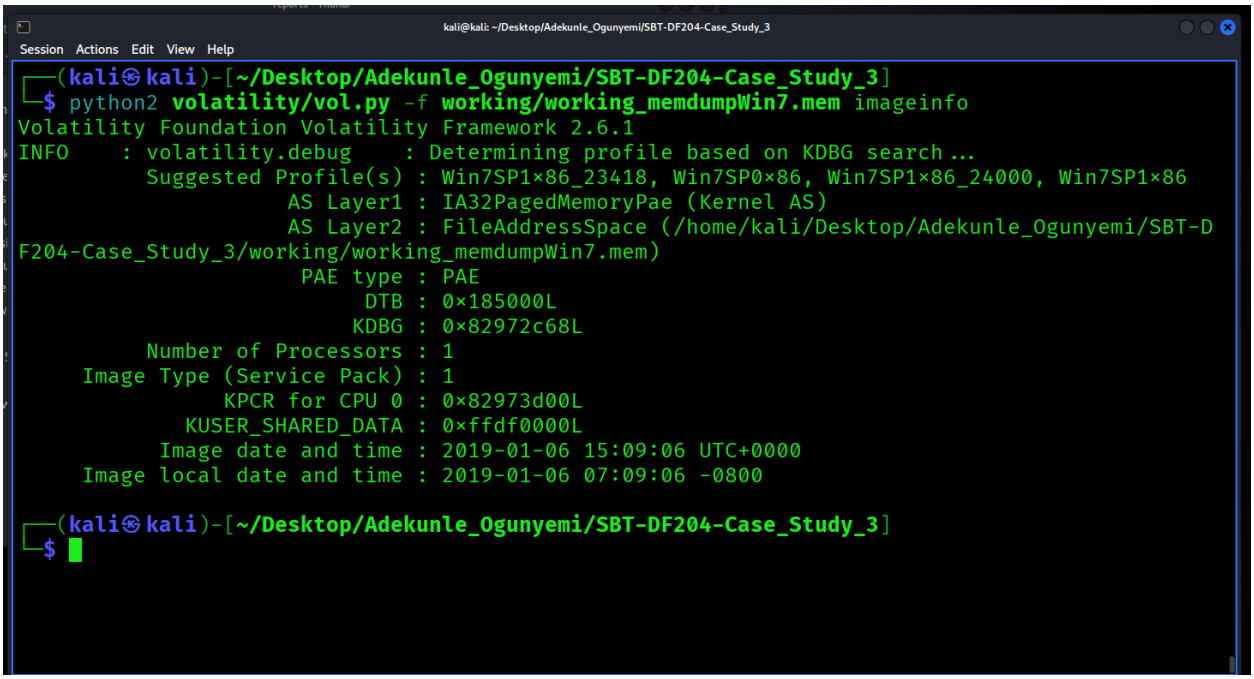


Figure 3: Terminal confirming the Volatility tool is available and ready for analysis

5 Volatility Analysis

The investigation was mainly based on command-line, registry, and File Explorer artefacts because the suspected transfer involved the Windows command prompt and a removable drive.

The following Volatility plugins were used during the investigation:

- imageinfo
- cmdscan
- consoles
- printkey -K "MountedDevices"
- shellbags

5.1 Finding 1 – Profile Identification

The first stage of the analysis was to identify the operating system profile supported by the memory image.

The following command was used:

python2 volatility/vol.py -f working/working_memdumpWin7.mem imageinfo

The results suggested the following profiles:

- Win7SP1x86_23418
- Win7SP0x86
- Win7SP1x86_24000
- Win7SP1x86

The system time was reported as 2019-01-06 15:09:06 UTC.

The profile Win7SP1x86 was tested using the pslist plugin. The output was successful, confirming this was the correct profile.

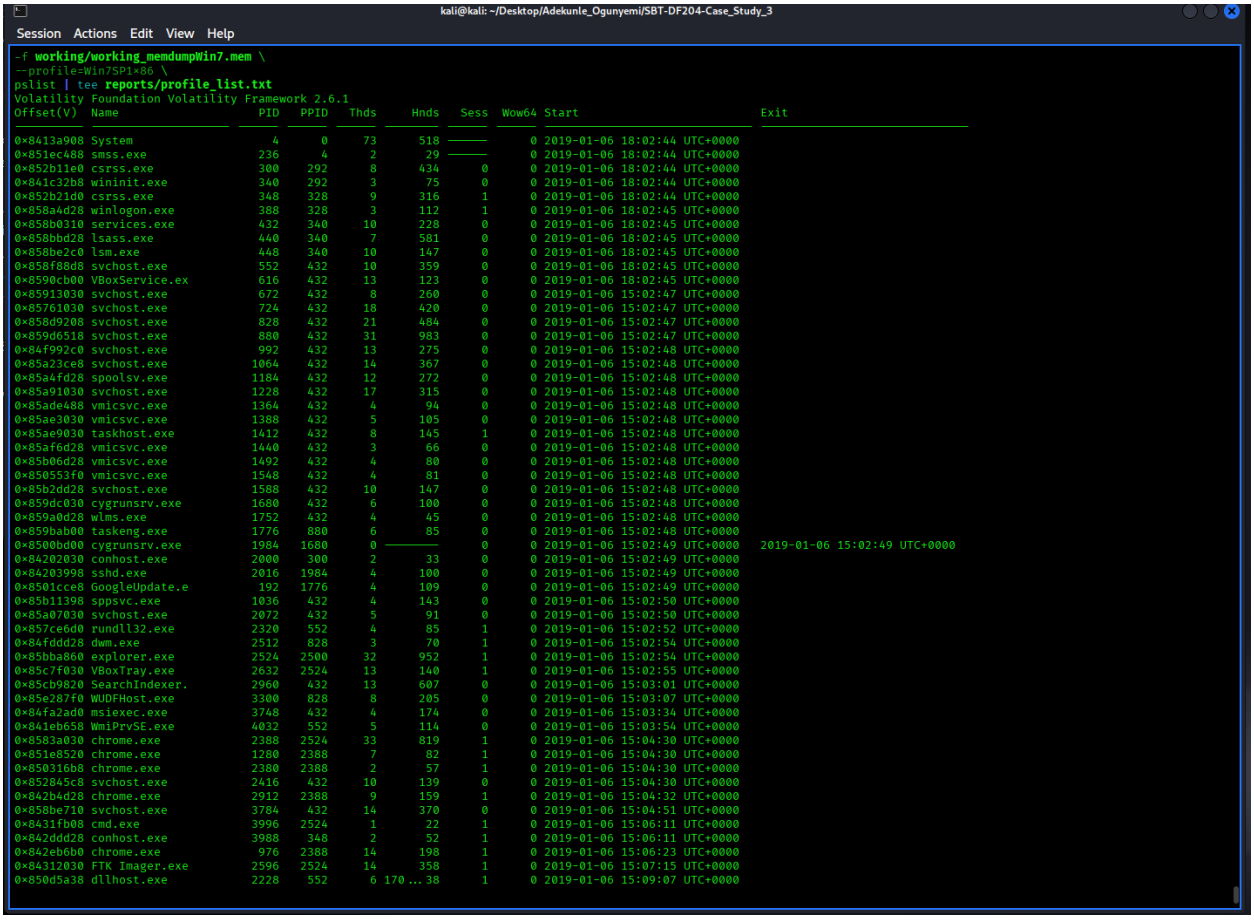


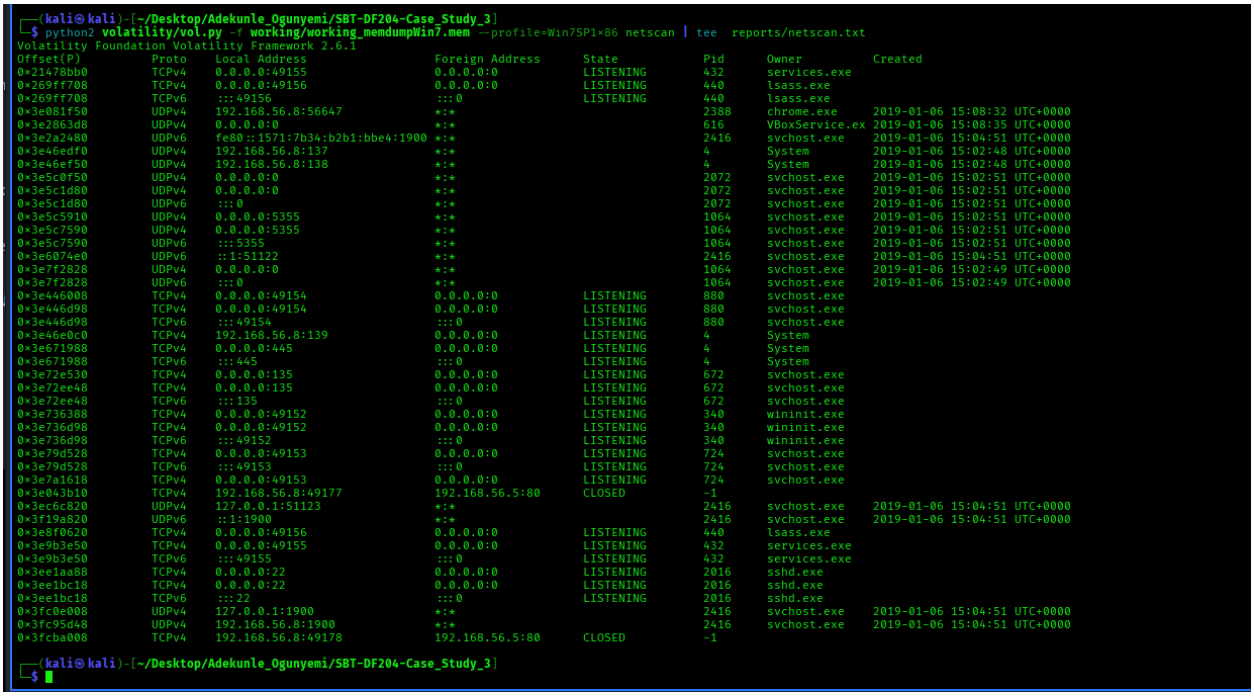
Figure 4: imageinfo output showing suggested Windows 7 profiles and system time

5.2 `Finding 2 – Process and Network Activity

After identifying the profile, I examined the running processes and network connections.

The netscan plugin identified a closed network connection from: Client IP: 192.168.56.8 to Destination IP: 192.168.56.5:80

The Process ID (PID) for this connection was -1, meaning it could not be linked to a specific running process.



```
┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF204-Case_Study_3]
└─$ python2 volatility/vol.py -f working/working_memdumpWin7.mem --profile=Win7SP1x86 netscan | tee  reports/netscan.txt
Volatility Foundation Volatility Framework 2.6.1
Offset(P)          Proto    Local Address                  Foreign Address      State            Pid      Owner          Created
0x21478bb0         TCPv4    0.0.0.0:49155                  0.0.0.0:0            LISTENING        432      services.exe
0x269ff708         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        440      lsass.exe
0x269ff708         TCPv6    :::49156                       :::0                 LISTENING        440      lsass.exe
0x3e081f50         UDPv4    192.168.56.8:56647             *:*                                   2388     chrome.exe     2019-01-06 15:08:32 UTC+0000
0x3e2863d8         UDPv4    0.0.0.0:0                      *:*                                   616      VBoxService.ex 2019-01-06 15:08:35 UTC+0000
0x3e2a2480         UDPv6    fe80::1571:7b34:b2b1:bbe4:1900 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e46edf0         UDPv4    192.168.56.8:137               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e46ef50         UDPv4    192.168.56.8:138               *:*                                   4        System         2019-01-06 15:02:48 UTC+0000
0x3e5c0f50         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv4    0.0.0.0:0                      *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c1d80         UDPv6    :::0                           *:*                                   2072     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c5910         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv4    0.0.0.0:5355                   *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e5c7590         UDPv6    :::5355                        *:*                                   1064     svchost.exe    2019-01-06 15:02:51 UTC+0000
0x3e6074e0         UDPv6    ::1:51122                      *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e7f2828         UDPv4    0.0.0.0:0                      *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e7f2828         UDPv6    :::0                           *:*                                   1064     svchost.exe    2019-01-06 15:02:49 UTC+0000
0x3e446008         TCPv4    0.0.0.0:49154                  0.0.0.0:0            LISTENING        880      svchost.exe
0x3e446d98         TCPv4    0.0.0.0:49154                  0.0.0.0:0            LISTENING        880      svchost.exe
0x3e446d98         TCPv6    :::49154                       :::0                 LISTENING        880      svchost.exe
0x3e46e0c0         TCPv4    192.168.56.8:139               0.0.0.0:0            LISTENING        4        System
0x3e671988         TCPv4    0.0.0.0:445                    0.0.0.0:0            LISTENING        4        System
0x3e671988         TCPv6    :::445                         :::0                 LISTENING        4        System
0x3e72e530         TCPv4    0.0.0.0:135                    0.0.0.0:0            LISTENING        672      svchost.exe
0x3e72ee48         TCPv4    0.0.0.0:135                    0.0.0.0:0            LISTENING        672      svchost.exe
0x3e72ee48         TCPv6    :::135                         :::0                 LISTENING        672      svchost.exe
0x3e736388         TCPv4    0.0.0.0:49152                  0.0.0.0:0            LISTENING        340      wininit.exe
0x3e736d98         TCPv4    0.0.0.0:49152                  0.0.0.0:0            LISTENING        340      wininit.exe
0x3e736d98         TCPv6    :::49152                       :::0                 LISTENING        340      wininit.exe
0x3e79d528         TCPv4    0.0.0.0:49153                  0.0.0.0:0            LISTENING        724      svchost.exe
0x3e79d528         TCPv6    :::49153                       :::0                 LISTENING        724      svchost.exe
0x3e7a1618         TCPv4    0.0.0.0:49153                  0.0.0.0:0            LISTENING        724      svchost.exe
0x3e043b10         TCPv4    192.168.56.8:49177             192.168.56.5:80      CLOSED           -1
0x3ec6c820         UDPv4    127.0.0.1:51123                *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3f19a820         UDPv6    ::1:1900                       *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3e8f0620         TCPv4    0.0.0.0:49156                  0.0.0.0:0            LISTENING        440      lsass.exe
0x3e9b3e50         TCPv4    0.0.0.0:49155                  0.0.0.0:0            LISTENING        432      services.exe
0x3e9b3e50         TCPv6    :::49155                       :::0                 LISTENING        432      services.exe
0x3ee1aa88         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv4    0.0.0.0:22                     0.0.0.0:0            LISTENING        2016     sshd.exe
0x3ee1bc18         TCPv6    :::22                          :::0                 LISTENING        2016     sshd.exe
0x3fc0e008         UDPv4    127.0.0.1:1900                 *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fc95d48         UDPv4    192.168.56.8:1900              *:*                                   2416     svchost.exe    2019-01-06 15:04:51 UTC+0000
0x3fcba008         TCPv4    192.168.56.8:49178             192.168.56.5:80      CLOSED           -1

┌──(kali㉿kali)-[~/Desktop/Adekunle_Ogunyemi/SBT-DF204-Case_Study_3]
└─$
```

Figure 5: netscan output showing a closed connection to 192.168.56.5 with no associated PID

5.3 Finding 3 – Command History

The cmdscan plugin was used to examine the command prompt's command history.

The results showed a command process running under cmd.exe (PID 3996) with the following commands:

- Command #0: ipconfig
- Command #1: cd Downloads

❖ Command #2: copy secret_file.docx F:

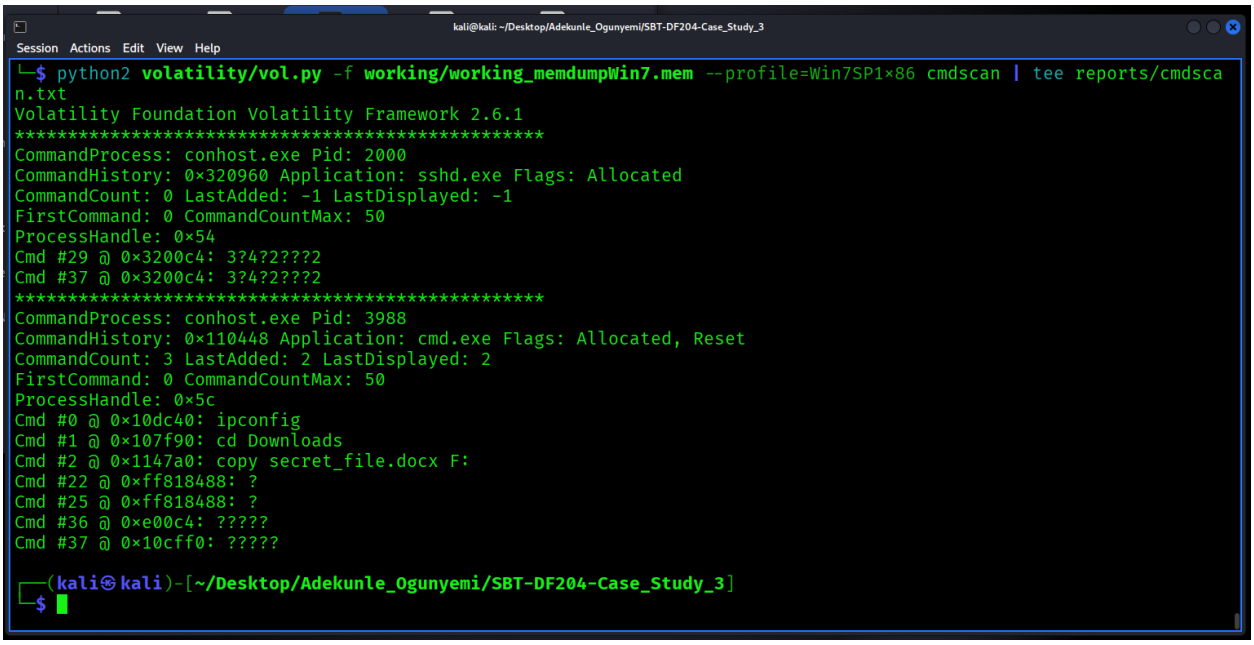


Figure 6: cmdscan output showing the command prompt history, including the copy command

5.4 Finding 4 – Console Output

To get the actual screen output of the command prompt, the consoles plugin was used.

The console dump showed the user navigating to C:\Users\IEUser\Downloads and then running the copy command. The system returned the message: "1 file(s) copied."

This confirmed that the command was executed successfully.

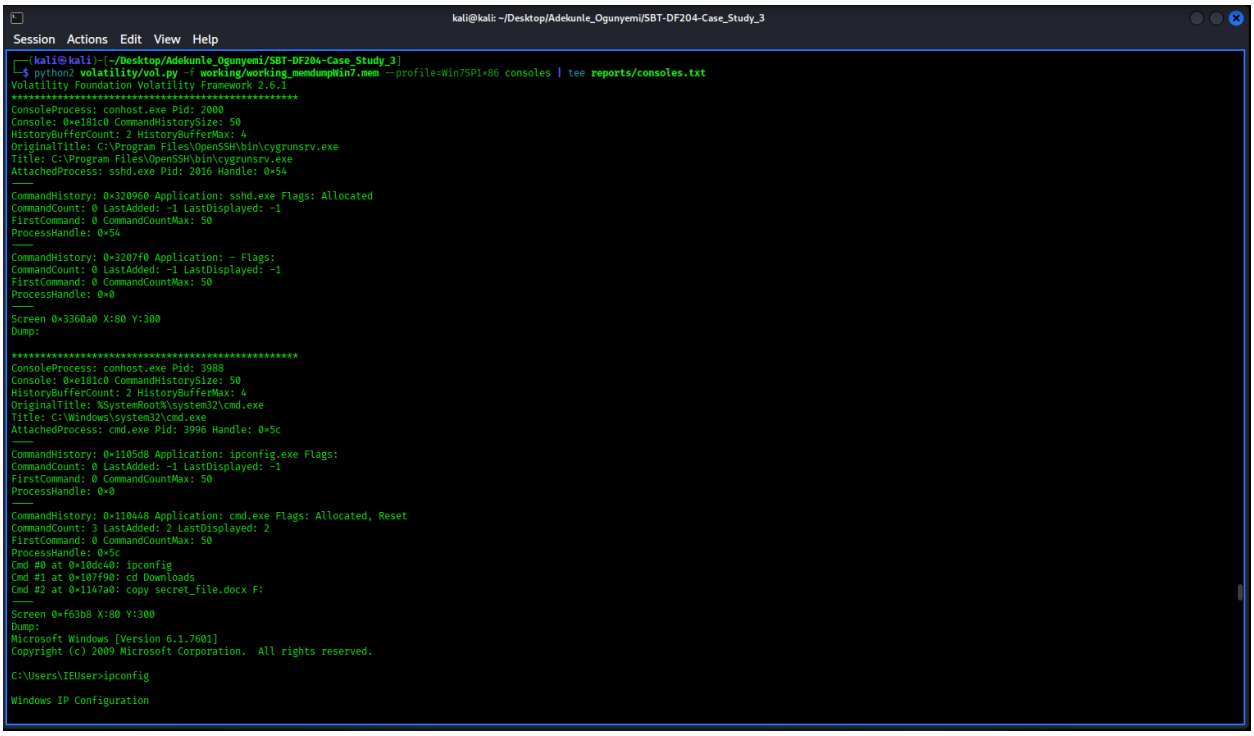


Figure 7: console output showing the command prompt screen and the "1 file(s) copied" message

5.5 Finding 5 – Removable Media Evidence

To determine what the F: drive was, the MountedDevices registry key was examined.

The output showed:
REG_BINARY \DosDevices\F: : (S)

The hex data for this key contained the string:
\??\USBSTOR#Disk&Ven_General&Prod_UDisk...

This proved that the F: drive was a physical USB Mass Storage Device (USB flash drive).

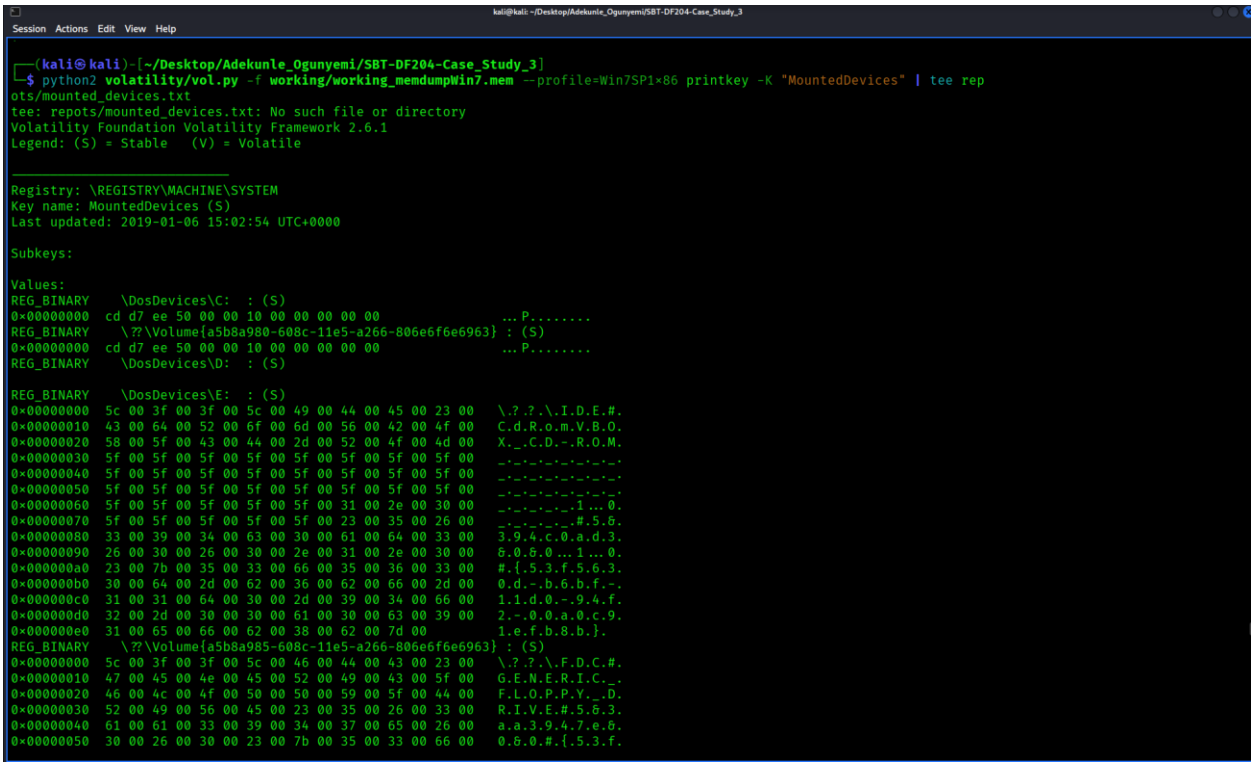


Figure 8: printkey output of the MountedDevices registry key showing F: is a USB device

5.6 Finding 6 – File Explorer History

The File Explorer history was examined using the shellbags plugin.

The results showed that the user had accessed the Downloads folder at 14:49:40 UTC:
C:\Users\IEUser\Downloads

The results also showed an entry for the F:\ drive, confirming that the drive was opened in Windows Explorer.

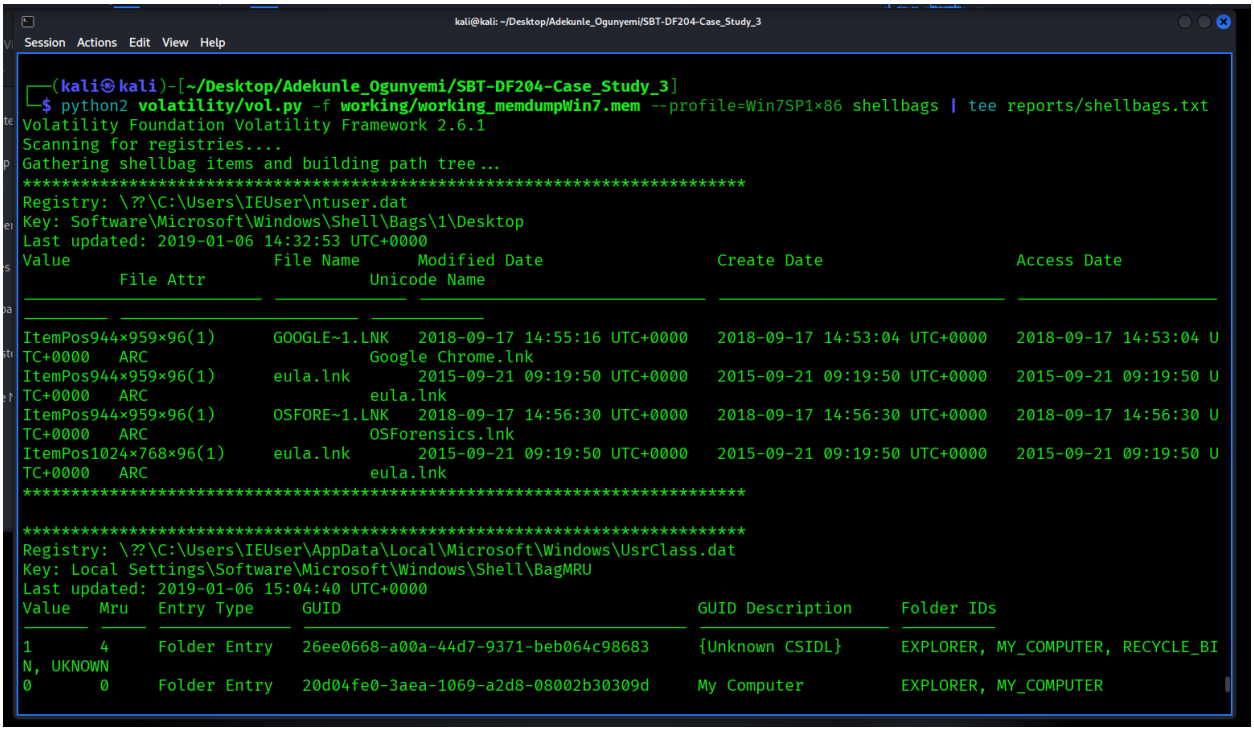


Figure 9: shellbags output showing the Downloads folder and F: drive in File Explorer history

6 Timeline of Relevant Activity

The important activity in the memory image can be arranged in the following order.

Stage	UTC Timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Browser or process evidence	15:08:32	E04	netscan	Closed connection to 192.168.56.5:80	Cannot link to the transfer (PID -1).
Download-folder review	14:49:40	E06	shellbags	User opened Downloads folder	Shows user was browsing files.
Removable-device evidence	15:02:54	E05	printkey	F: is a USBSTOR device	Proves the destination was a USB drive.
Drive mapping	15:07:03	E06	shellbags	F:\ volume accessed	Confirms user opened the USB drive.

Command-history evidence	~15:06	E03	cmdscan	copy secret_file.docx F:	Direct evidence of the copy command.
Supported conclusion	~15:06	E03/E04	consoles	"1 file(s) copied"	Proves the transfer was successful.

The relevant transfer occurred at approximately 15:06 UTC.

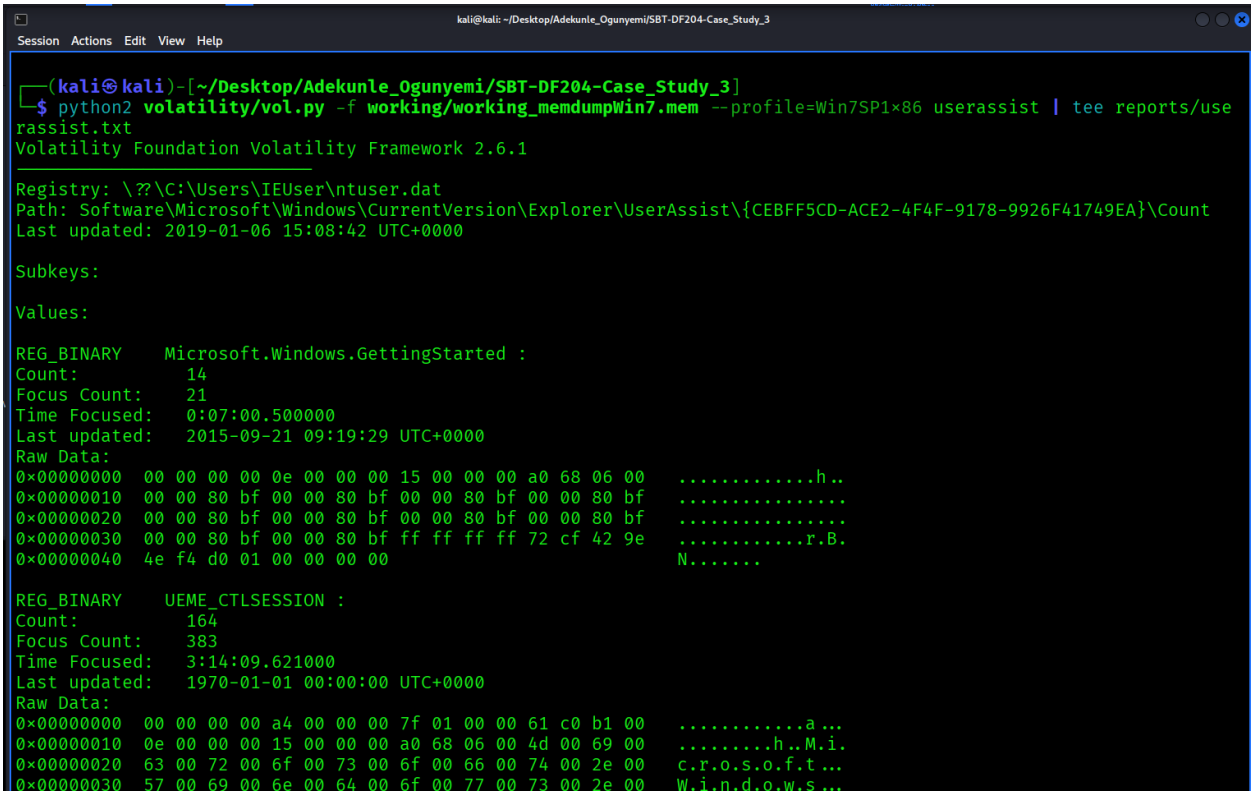


Figure 10: Terminal showing the exact sequence of Volatility commands used to build the timeline

7 Evidence Log

Evidence ID	Evidence / Plugin	Finding	Why it matters	Screenshot
E01	sha256sum	Original and working copy have matching hashes	Shows integrity of the working copy	Figure 2

E02	imageinfo	Win7SP1x86 profile, 15:09:06 UTC	Establishes the correct profile and image time	Figure 4
E03	cmdscan	copy secret_file.docx F:	Directly connects the user to the copy command	Figure 6
E04	consoles	"1 file(s) copied"	Proves the copy operation was successful	Figure 7
E05	printkey	\DosDevices\F: is a USBSTOR device	Identifies the destination as a USB drive	Figure 8
E06	shellbags	Downloads folder and F: drive accessed	Corroborates the sequence of activity	Figure 9
E07	netscan	Closed connection to 192.168.56.5 (PID -1)	Shows network activity but cannot link it	Figure 5

8 Attribution Assessment

The evidence can be divided into three levels.

8.1 Direct Evidence

The direct evidence shows that:

- The active user account was IEUser.
- The user ran ipconfig.
- The user ran cd Downloads.
- The user ran copy secret_file.docx F:.
- The console confirmed, "1 file(s) copied."
- The F: drive was a USB Mass Storage Device.

8.2 Reasonable Inference

From the evidence, it is reasonable to conclude that the IEUser account was used to copy secret_file.docx to a USB drive.

The ShellBags history supports this by showing that the Downloads folder and the F: drive were browsed through Windows Explorer.

8.3 What the Evidence Does Not Prove

The evidence does not prove with absolute certainty that a specific physical person was sitting at the computer and typing the commands at the exact moment they were run.

The IEUser account could have been shared, or left logged in while another person used the computer.

For this reason, the report should not state that the evidence alone proves the identity of the person who performed the transfer.

9 Limitations of the Investigation

9.1 Shared User Account

The evidence identifies the IEUser account, not a specific person. This is a significant limitation when making an attribution.

9.2 Device Does Not Equal Person

The evidence proves the file was copied to a USB drive. It does not prove who plugged in the USB drive or who removed it afterwards.

9.3 Network Anomaly

The netscan output showed a closed connection to 192.168.56.5. However, because the PID was -1, this connection cannot be linked to the cmd.exe process or the file transfer. It must be noted as an alternative explanation that could not be fully investigated.

10 Conclusion

The investigation of memdumpWin7.mem identified an active user account named IEUser. The command history and console output showed that the user navigated to the Downloads folder and ran the command:

copy secret_file.docx F:

The console output confirmed the operation was successful with the message "1 file(s) copied."

The MountedDevices registry key proved that the F: drive was a physical USB Mass Storage Device. The ShellBags history confirmed that the user had browsed the Downloads folder and the F: drive.

Based on these findings, the evidence strongly supports that the IEUser account was used to copy the confidential file secret_file.docx to a USB drive.

However, I would not describe the evidence as absolute proof that a specific person physically operated the computer when the transfer occurred. The shared nature of user accounts and the inability to link the USB device to a specific individual are important limitations.

Overall, the evidence provides a strong connection between the user account, the command prompt activity, and the USB device, but the final attribution should be stated with the limitations described above.

11 Recommendation

For stronger attribution, additional evidence could be examined if it were available. This could include:

- Windows Event Logs for logon and logoff activity.
- USB device connection and disconnection times.

- ❖ Physical access records for the workstation.
- ❖ CCTV footage if available.
- ❖ Examination of the USB device itself.
- ❖ Network logs to investigate the 192.168.56.5 connection.

These additional sources could help establish who was actually using the device at the time of the transfer.

12 Final Statement

This investigation was carried out using the supplied memory image and Volatility 2.6.1. The original evidence was preserved, a working copy was created, and the hash values were compared before analysis.

The memory image provided enough information to trace the unauthorised copy of secret_file.docx to the IEUser account and to prove that the destination was a USB drive.

The evidence therefore provides a strong association between the IEUser account and the USB transfer of the confidential document, while the shared nature of user accounts remains a limitation when making a final statement about the person who physically performed the transfer.

Author
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